THE UNIVERSITY OF NOTRE DAME AUSTRALIA

MED 4000

WEEK 22

TUTOR GUIDE

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- Members of the University of Notre Dame clinical year's curriculum committee [CYC]. Last Reviewed , October 2010

RELEVANT PRIOR LEARNING:

MED 1000

1.11 Pathophysiology of diabetes. [L]
1.32 Anatomy of the eye.

MED 2000

2.03 History & examination of the eye [L & CCS]
2.04 Autoimmunity [L]
2.26 Assessment & management of the diabetic patient [L & CCS]
2.26 Diabetes whole body changes [L]
2.30 History & examination in thyroid disease [L]
2.30 Assessment of patients with thyroid disease

MED 3000

3.16 Diabetes – ongoing management [Short case tutorial & CCS]
3.22 Rheumatoid disease [Short case tutorial]

WEEK 22_SURGERY_OPHTHALMOLOGY_MEDICINE		
Yr	**Wk**	**LEARNING OBJECTIVES:**
4	**22**	• Define glaucoma; know the commonest form and the key differences between open angle and closed angle glaucoma. • Outline the causes of vision loss. • Briefly describe the medical and surgical methods used in treatment of normal tension open angle glaucoma. • List the key symptoms and signs which would indicate acute angle closure glaucoma, which is an ophthalmic emergency.
4	**22**	• Know the two distinct mechanisms by which diabetes mellitus causes vision loss and blindness. • List the risk factors which will indicate a greater need to screen for retinopathy in diabetic patients. • Outline the frequency of retinal screening in diabetic patients including which health professionals can perform them. • Briefly describe the uses of photocoagulation in the treatment of patients with proliferative diabetic retinopathy.
4	**22**	• List the eye manifestations of thyroid disease. • Outline the factors contributing to dry eyes. • Know how keratoconjunctivitis sicca (dry eyes) is diagnosed. • Briefly describe the treatments available for keratoconjunctivities sicca and the other manifestations of thyroid eye disease.
4	**22**	• List the four possible eye diagnoses which are linked to rheumatoid arthritis. • Know the difference between scleritis and episcleritis in presentation and outcome.

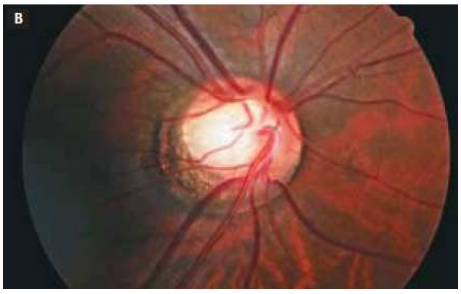
CASE ONE

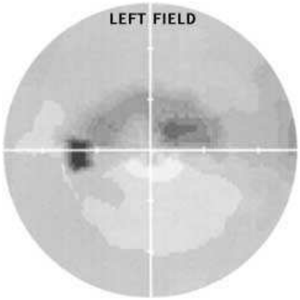
Short case number: 4_23_1
Category: Nervous system & Eye
Discipline: Medicine
Setting: General practice
Topic: Acute and chronic glaucoma

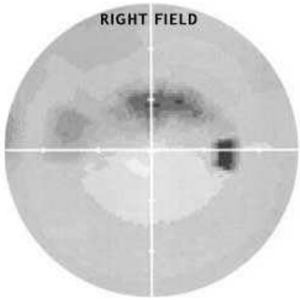
CASE

Margaret Fong is a 58 year old retired teacher who initially presented to her optometrist for new glasses following a near accident. The optometrist's referral letter described bilateral optic disc cupping with cup-to-disc ratio of 70% and Intra-occular pressure readings R: 18 and L: 17 mmHg [Normal 10-20 mmHg].

Significant history includes presbyopia for which she wears reading glasses. Margaret also has mild hypertension which is controlled on an ACE inhibitor. Nil other medications. No family history of glaucoma or diabetes.







On current examination, BP was 140/90 mmHg and indirect fundoscopy confirmed bilateral pronounced optic disc cupping. Visual field testing revealed arc-shaped blind areas.

QUESTIONS

1. Margaret wants to know what's wrong with her eyes now. Outline how you would explain to her what glaucoma is, and what is the difference between open angle and closed angle glaucoma?
2. The patient asks what treatments are available for normal tension (open angle) glaucoma. Outline the treatments used.
3. What type of presentation would indicate acute angle-closure glaucoma, which is an ophthalmic emergency?

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 21 Special senses p 1091-2. Saunders Elsevier 2009.
- Graham SL. How to treat: Glaucoma. Australian Doctor, March 2007, p 23-30.
- NHMRC Guidelines for the screening, prognosis, diagnosis, management and prevention of glaucoma 2010. http://www.nhmrc.gov.au/publications/synopses/cp113syn.htm

ANSWERS

Question 1
Margaret wants to know what's wrong with her eyes now. Outline how you would explain to her what glaucoma is, and the difference between open angle and closed angle glaucoma?

GLAUCOMA is a progressive optic neuropathy characterised by the combination of both changes in the structure of the optic nerve and loss of visual field. There are two main types of glaucoma: open angle and closed angle.

The main risk factors for glaucoma are:
• AGE: People at any age, however the level of risk significantly increases for those > 50 years
• People with a FAMILY HISTORY of glaucoma
• People with diabetes
• People with myopia/short sightedness
• People who have a history of migraine
• People taking some medications for a range of health conditions

1. By far the most common form is **primary open-angle glaucoma (POAG). Margaret has normal-tension glaucoma**, which is a subgroup of the POAG. It is presumed that patients with this condition have nerves that are highly susceptible to pressure, a vascular pathophysiology or some other form of neurodegenerative disease. They still show a clinical response to lowering of IOP, so they are treated as for POAG.

2. **Angle-closure/Closed angle glaucoma** is much less common, but has a higher incidence in Asian people. It involves quite a different mechanism and presentation. The angle is the region between the cornea and the base of the iris, where the drainage channels are located (trabecular meshwork).

Symptoms described by the patient:
Open angle glaucoma is generally symptomless in its early stages. It is not until significant neuronal damage has occurred that characteristic visual loss is observed.
Acute angle closure glaucoma is associated with significant and distressing symptoms. These may present as either an acute scenario, or as patient descriptions of past attacks.
Chronic angle closure glaucoma symptoms are often absent. The symptoms that should alert health care providers to the presence of AC are detailed in Table 7.1, extracted from the European Glaucoma Society [EGS] (2003).

Table 1: The two main types of glaucoma		
Features	**Open angle**	**Closed angle**
Presenting symptoms	None usually, central vision loss only in late stages, unrecognised mid-peripheral field loss progresses to tunnel vision eventually	Blurring, haloes, pain – deep ache, redness, headache, vomiting
Signs	Disc cupping, visual field loss, raised intraocular pressure (IOP)	Mid-dilated non-reacting pupil, very high IOP, cloudy cornea (figure 1)
Referral	Not urgent — very slow disease process	Emergency — rapid visual loss can follow
Treatment	Control IOP with topical therapy, laser or surgery	Immediate IOP lowering, topical IV, urgent laser treatment to open angle
Prognosis	Very good if detected early and managed well. About 10% still show slow progression	Good if prompt treatment given, but many still require treatment and surgery later
Differential diagnosis	Exclude optic nerve compression and tumours if atypical features	Iritis, keratitis, herpes zoster, ophthalmicus, scleritis

Up to 50% of glaucoma in the community remain undiagnosed because the detection of POAG can be elusive until the disease process is well advanced. A substantial amount of vision can be lost before the patient becomes aware of any defect.

Diagnosis of open angle glaucoma (OAG) must be based on multiple sources of information. This condition can occur in eyes with normal or raised IOP. The concept that POAG only occurs with pressures over 21mmHg is outdated. However, whilst increasing emphasis is now placed upon the morphological changes occurring at the optic nerve head and retinal nerve fibre layer, IOP measurement remains core to any diagnosis and management of glaucoma.

Question 2
The patient asks what treatments are available for normal tension (open angle) glaucoma. Outline the treatments used.

a) Medications: Normal-tension glaucoma will still show a clinical response to lowering of intraocular pressure, so they are treated as for POAG. Regarding side effects, the main risk is that of combining systemic beta blockers with topical preparations, increasing the risk of side effects, especially now there are several combination products containing timolol. Many products contain warnings for use in glaucoma (e.g. anticholinergics, tricyclics antidepressants). This almost always refers to patients with narrow angles, who are at risk of angle closure, and is not a factor in POAG patients.

Table 3: Medications used in glaucoma, and their main side effects		
Class	**Generic/trade names**	**Side effects**
Beta blockers	Timolol (0.25%, 0.5%) (Nyogel 0.1%, Tenopt, Timoptol, Timoptol XE, Timoptic), Levobunolol 0.25% (Betagan), Betaxolol 0.25%, 0.5% (Betoptic)	Bronchoconstriction, bradycardia, fatigue, headache, confusion, depression, nightmares, impotence, effects on lipids, masking of hypoglycaemia
Prostaglandin derivatives	Latanoprost 0.005% (Xalatan), Travoprost 0.004% (Travatan), Bimatoprost 0.03% (Lumigan)	Iris colour change to brown, eyelash growth, periorbital pigment, hyperaemia, HSV keratitis reactivation
Alpha$_2$ agonists	Aproclonidine 0.5% (Iopidine), Brimonidine 0.2%, (Alphagan)	Local allergy, fatigue, dry mouth, blurring
Miotics	Pilocarpine 0.5-6%	Miosis, headache, blurring, reduced night vision
Carbonic anhydrase inhibitors		
Topical	Dorzolamide 2 % (Trusopt), Brinzolamide 1% (Azopt)	Local irritation, metallic taste
Oral	Acetazolamide 250mg (Diamox)	Parasthesiae, malaise, fatigue, depression, hypokalaemia, metabolic acidosis, gastrointestinal disturbances, kidney stones

b) Laser therapy

In POAG, laser trabeculoplasty is usually used as a second-line treatment when eye drops have either not provided sufficient effect on their own, or are not well tolerated because of side effects. The most commonly used laser therapy, the argon laser, is used for a procedure known as an argon laser trabeculoplasty. A second type of treatment is the peripheral laser iridotomy, which uses the laser to create a channel through the iris. The purpose of the channel is to prevent or treat angle-closure glaucoma.

Question 3

What type of presentation would indicate acute angle-closure glaucoma (AACG), which is an ophthalmic emergency?

Table 7.1: Symptoms of angle closure

SYMPTOMS	Acute angle closure	Intermittent angle closure	Chronic angle closure
Blurred vision	√	At time of attack presents as AAC Between attacks may be symptomless	Variable – chronic angle closure mimics primary open angle glaucoma It is asymptomatic until visual field loss interferes with quality of life
Coloured rings around lights	√		Transient if present
Pain	√		Not usually
Frontal headache	√		Discomfort rather than pain
Palpitations and abdominal pain	√		×
Nausea and vomiting	√		×

In this type of glaucoma there is a sudden rise in intraocular pressure to levels > 50 mmHg. This occurs due to reduced aqueous drainage as a result of the ageing lens pushing the iris forward against the trabecular meshwork. People most at risk of developing AACG are those with shallow anterior chambers such as hypermetropes and women. The attack is more likely to occur under reduced light conditions when the pupil is dilated.

AACG causes sudden onset of a red painful eye and blurred vision. Patients become unwell with nausea and vomiting and complain of headache and severe ocular pain. The eye is injected, tender and feels hard. The cornea is hazy and the pupil is semi-dilated. Emergency Box 20.1 (above) shows features that require urgent referral to an ophthalmologist.

Prompt treatment is required to preserve sight and includes i.v. acetazolamide 500 mg (provided there are no contraindications) to reduce IOP, and instillation of pilocarpine 4% drops to constrict the pupil to improve aqueous outflow and prevent iris adhesion to the trabecular meshwork. Other topical drops such as beta-blockers and prostaglandin analogues can also be instilled if available, provided there are no contraindications. Analgesia and antiemetics are given as required.

These patients must be referred to an ophthalmologist immediately so that reduction in IOP can be monitored and other agents such as oral glycerol or i.v. mannitol can be administered to non-responding patients. Definitive treatment involves making a hole in the periphery of the iris of both eyes either by laser or surgically.

CASE TWO

Short case number: 4_23_2
Category: Nervous system & Eye
Discipline: Medicine
Setting: General practice
Topic: Systemic disease eye changes_diabetes

CASE

Beverley Higgs [3_16_4] is a 59 year old patient with poorly controlled diabetes who previously presented with deteriorating vision. She had not been attending the surgery for 6 years since she was staying with her daughter, but on that day she was attending for a flu shot. When asked directly, Beverley admitted that her vision had deteriorated in the last couple of years.

As part of the GP Management Plan she had been referred to an ophthalmologist for initial screening. At that time there was no sign of diabetic retinopathy or other eye disease. However Beverley did not attend for follow up, was inconsistent with her medications and found it very difficult to manage her diet, or to exercise.

On this visit you noted the presence of proliferative retinopathy. You decided to have a discussion with Beverley about the necessity for referral to the ophthalmologist for laser photocoagulation.

QUESTIONS

1. What is the most important message to put to Beverley to convince her to return to the ophthalmologist for a consultation?
2. Regular screening for retinopathy is essential in all diabetic patients but is particularly important in those with risk factors. What are these risk factors?
3. How often should retinal screening occur in diabetic patients and who can perform the screening?
4. Outline the uses of photocoagulation in the treatment of a patient with proliferative retinopathy, like Beverley.

Resources:

- **Frier BM and Fisher M, Chapter 21 Diabetes Mellitus, pp 838-41, in Davidson's principles and practice of Medicine 20th ed. Churchill Livingstone, Elsevier 2006.**
- **Spurling G, Askew D and Jackson C. Retinopathy, screening recommendations. Australian Family Physician, October 2009, Vol. 38, No. 10, p 780-3.**

ANSWERS

Question 1
What is the most important message to put to Beverley to convince her to return to the ophthalmologist for a consultation?

Diabetic retinopathy (DR) is the leading cause of preventable blindness in adults worldwide, but early detection and appropriate treatment can prevent nearly all severe vision loss and blindness. Treatment to prevent visual loss will ultimately be needed by 30% of those with T2DM and 50% of those with type 1 diabetes.

Advanced DR causes vision loss and blindness via two distinct mechanisms. The most common, diabetic macular oedema, occurs as the blood retinal barrier breaks down with subsequent blood vessel leakage and retinal thickening. The other mechanism is via proliferative diabetic retinopathy (PDR) where neovascularisation occurs in response to ischaemia. This results in bleeding of new and abnormal vessels. Visual loss is associated with vitreal haemorrhage or retinal detachment. Laser photocoagulation remains the mainstay of treatment with reductions in severe visual loss by at least 50%. Vision threatening disease owing to PDR or diabetic maculopathy was present in 1.2% of the AusDiab study population with diabetes.

In the Australian population, at least 22% of Australians with diabetes have not seen either an ophthalmologist or an optometrist within the 2 year timeframe recommended by the guidelines. There are clearly many people who do not access appropriate screening. Furthermore, people with DR were no more likely to have had a retinal examination than those without DR, despite being at increased risk of severe visual loss or blindness.

Question 2
Regular screening for retinopathy is essential in all diabetic patients but is particularly important in those with risk factors. What are these risk factors?

Regular screening for retinopathy is essential in all diabetic patients but is particularly important in those with **risk factors**. These include

- early onset
- long duration of diabetes
- hypertension
- poor glycaemic control
- pregnancy
- use of the oral contraceptive pill
- smoking
- excessive alcohol consumption and
- evidence of microangiopathy elsewhere, particularly neuropathy and proteinuria.

Question 3
How often should retinal screening occur in diabetic patients and who can perform the screening?
The general recommendation is that GPs need to ensure that their patients with diabetes have been appropriately screened with a dilated fundus and a trained examiner <u>every 2 years</u>. However, many patients with diabetes will have an extra risk factor necessitating yearly screening

as per National Health and Medical Research Council (NHMRC) recommendations. This can be provided by suitably trained GPs, optometrists or where available, ophthalmologists. Some doctors remain concerned about mydriatic drops owing to risks of acute angle closure glaucoma. However, at a rate of 1–6 per 20 000 people, this is uncommon and has not been known to occur with 0.5% or 1% tropicamide drops, which are commonly used in Australian general practice.

a) Many GPs refer their patients to a private **ophthalmologist** or the local public hospital ophthalmology outpatient department for diabetic retinopathy screening. Ophthalmologists generally prefer biomicroscopy of a dilated eye with a special lens in conjunction with a slit lamp. Direct ophthalmoscopy even in the hands of ophthalmologists, is a less effective screening tool owing to limitations of the instrument itself.
b) **Optometrists** are another option and have been shown in a number of studies to have safe levels of sensitivity and specificity for detecting diabetic retinopathy internationally6 and within Australia. Appropriately trained GPs and physicians have been shown to safely detect DR. This can be achieved using retinal photography in primary care and has been successfully used in the Australian Indigenous health context. Tele-ophthalmology in conjunction with retinal photography is another option, especially for rural GPs working where access to an ophthalmologist is difficult. This usually involves the transfer of a digital image of the retina for assessment by a distant ophthalmologist. While prevention of DR through tight sugar control and blood pressure control are reasonable goals, GPs need to focus on ensuring their patients with diabetes access timely and appropriate screening and treatment with laser photocoagulation to prevent visual loss.
c) Screening should be undertaken by trained personnel in an organised and audited programme. The preferred screening option is a **digital imaging photographic system**, with reference if necessary to an ophthalmologist examination by slit lamp biomicroscopy.

Question 4
Outline the uses of photocoagulation in the treatment of a patient with proliferative retinopathy, like Beverley.

Severe non-proliferative and proliferative retinopathy is treated with retinal photocoagulation, which has been shown to reduce severe visual loss by 85%. **Photocoagulation is used:**

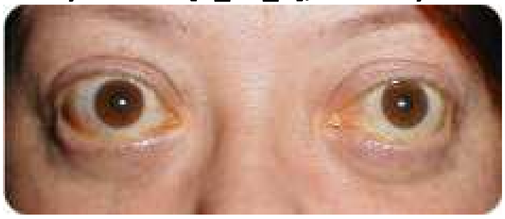
- **to destroy areas of retinal ischaemia (since it is thought that this plays a major role in the development of neovascularisation)**
- **to seal leaking microaneurysms and reduce macular oedema**
- **to gliose new vessels directly on the retinal surface (but not on the optic disc).**

Argon laser photocoagulation is the usual form of laser used for pan-retinal photocoagulation and for treatment of macular oedema. This simple procedure can be carried out with topical anaesthesia and in skilled hands carries little risk; it can be very effective. Pan-retinal photocoagulation results in new vessel elimination, with vision being maintained in up to 90% of patients who have new vessels on the retina and/or disc; macular oedema is also successfully treated in many patients with focal laser therapy. Patients must be reviewed regularly to check for further development of new vessels and/or maculopathy. Extensive bilateral photocoagulation can cause significant visual field loss, which may interfere with driving ability and reduce night vision.

Vitrectomy may be used in selected cases with advanced diabetic eye disease where visual loss is due to recurrent vitreous haemorrhage which has failed to clear or retinal detachment resulting from retinitis proliferans.

CASE THREE

Short case number: 4_23_3
Category: Nervous system & Eye
Discipline: Medicine
Setting: Hospital endocrinology department
Topic: Systemic disease eye changes_thyroid disease

CASE
Shelley Hunter [3_17_2], the 35 year old with Graves disease, presents 3 months after treatment with radio-iodine with a constant irritated feeling in her eyes. This was getting her down because it was affecting her sleep. Moreover the problem with her 'startled look' was getting worse to the point where she could no longer wear eye make-up and felt too embarrassed to go out. She was aware that she has been treated for Graves disease and hyperthyroidism. Her GP had prescribed a tricyclic antidepressant because she had been very depressed about her diagnosis. What she wanted to know was why she was getting worse and whether the eye problems were related with the thyroid.

On examination, there was no tremor or tachycardia. Bilateral conjunctival oedema and inflammation was present. Although there was noticeable proptosis, lid lag was minimal and Shelley was able to close her eyes.

QUESTIONS
1. How would you explain to Shelley the connection between her eye symptoms and thyroid disease? 2. What are the factors that contribute to dry eyes? How is kerato-conjunctivitis sicca diagnosed? 3. What treatment options are available to Shelley for symptomatic relief? 4. What differential diagnoses should be considered in a patient presenting with proptosis?

Resources:

- **Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 18, Endocrine disease pp 991-2. Saunders Elsevier 2009.**
- **Hodge C and Martin P. Eye series 10: Thyroid eye disease. Australian Family Physician, November 2003, Vol. 32, No. 11, p 939-40.**
- **Hodge C and Martin P. Eye series 3: Dry eyes. Australian Family Physician, April 2003, Vol. 32, No. 4, p 1-2.**

ANSWERS

Question 1
How would you explain to Shelley the connection between her eye symptoms and thyroid disease?

Eye disease is a manifestation of Graves' disease and can occur in patients who may be hyperthyroid, euthyroid or hypothyroid. Thyroid dysfunction and ophthalmopathy usually occur within 2 years of each other although sometimes a gap of many years is seen. Ophthalmopathy is more common and more severe in smokers.

The ophthalmopathy of Graves' disease is due to a specific immune response that causes retro-orbital inflammation. Swelling and oedema of the extraocular muscles lead to **limitation of movement** and to **proptosis** which is usually bilateral but can sometimes be unilateral. Ultimately increased pressure on the optic nerve may cause optic atrophy.

A **high proportion of patients with Graves' disease notice some soreness, painful watering or prominence of the eyes, and the 'stare' of lid retraction is relatively common**. More severe proptosis occurs in a minority of cases, and limitation and discomfort of eye movement and visual impairment due to optic nerve compression are relatively uncommon. Proptosis and lid retraction may limit the ability to close the eyes completely so that corneal damage may occur. There is periorbital oedema and conjunctival oedema and inflammation.

Eye manifestations do not parallel the degree of biochemical thyrotoxicosis, or the need for antithyroid therapy, but **exacerbation of eye disease is more common after radioiodine treatment** (15% vs 3% on antithyroid drugs). Sight is threatened in only 5–10% of cases, but the discomfort and cosmetic problems cause great patient anxiety.

Question 2
What are the factors that contribute to dry eyes? How is kerato-conjunctivitis sicca diagnosed?

Dry eye is typically associated with a deficiency in aqueous tear production, known as *kerato-conjunctivitis sicca*. It generally arises from a deficiency or compromise to the quality or quantity of the 3 layers of the tear film (out oily/lipid, middle/aqueous, innermost mucous/mucin layer). Patients often complain of bilateral constant and disabling eye burning, foreign-body sensation, redness or having "tired" eyes.

Factors that may influence this include:

- **Age**: Almost 75% of the population over 65 years of age will experience dry eye symptoms. As we age decreased production of the aqueous occurs, similarly the body produces less oil as we grow older. More pronounced in women, this oil deficiency allows the tear film to evaporate more quickly leading to further irritation.
- **Medications**: Many medications can impair lacrimal gland function leading to dry eyes. These include antibiotics, antihistamines, diuretics, oral contraceptives and antianxiety medications. Drugs such as tricyclic antidepressants, beta blockers and antihistamines can also exacerbate dryness. Dry eyes are common in postmenopausal women and can be exacerbated by HRT.

- **Inflammation of the eyelids or skin**: Other causes of drying such as meibomian gland dysfunction from blepharitis, or mechanical eyelid problems such as ectropion causing corneal exposure, must also be considered.
- **Systemic disease**: Dry eye may also be associated with systemic immune dysfunction from diseases such as rheumatoid arthritis and Sjögren's syndrome. Patients with Sjögren's syndrome have concurrent dry mouth symptoms.
- Localised irritation or irregularities of the ocular surface, and
- Previous eye surgery.

Diagnosis of keratoconjunctivitis sicca: Signs of dryness include a decreased tear film height or increased mucus production in the inferior fornix. Tear film height is best evaluated at a slit lamp, where the tear meniscus normally measures 0.5mm above the lower eyelid margin. Punctate staining of the conjunctiva and the cornea in the interpalpebral zone with either Rose Bengal or fluorescein dye can be commonly seen.

Treatment: A severe dry mouth or systemic signs of rheumatological disease should prompt investigation of serological inflammatory markers to look for systemic disease. Patient who are refractory to treatment and require excessive amounts of lubricants should be referred. Temporary or permanent occlusion of the lacrimal puncti is often performed by an ophthalmologist, who will also perform tests to quantify tear production.

Question 3
What treatment options are available to Shelley for symptomatic relief?

If the patient is thyrotoxic this should be treated, but this will not directly result in an improvement of the ophthalmopathy, and hypothyroidism must be avoided as it may exacerbate the eye problem. Smoking should be stopped.

Thyroid eye disease is a self limited process. The condition has 2 main phases. The first stage is the acute stage characterised by the onset of the inflammation and subsequent ocular symptoms. This is followed by a more stable quiescent phase. Although inflammation has usually subsided by this stage, fibrosis of the tissue can occur. The process can last between 12–24 months. With appropriate treatment it is rare that vision is severely affected.

Treatment of the eyes may be either local or systemic, and always requires close liaison between specialist endocrinologist and ophthalmologist:
■ Tear supplements are available as drops, ointments or gels and should be started. Most older lubricating drops (Murine or Tears Naturale) contain preservatives that can be toxic to the ocular surface, particularly with often and prolonged use. Newer products with less toxic preservatives (GenTeal or Polytears), unpreserved Murine or Tears Naturale, or single-dose lubricants (Cellufresh or Celluvisc) are recommended if prescribed more than four times a day.
■ Medications that contribute to dryness should be discontinued and patients should be counselled to avoid dry or windy environments.
■ The eyelids can be taped to ensure closure at night. In the rare case, a tarsorrhaphy (eyelids partly sewn together) is required.
■ Systemic steroids usually reduce inflammation if more severe symptoms are present.
■ Lid surgery will protect the cornea if lids cannot be closed.

■ Corrective eye muscle surgery may improve diplopia due to muscle changes, but should be deferred until the situation has been stable for 6 months and should follow any orbital decompression.
■ Unless optic nerve compression or severe corneal disease has occurred, surgery should be delayed until the later quiescent phase of thyroid eye disease. In cases of optic nerve compression, if no response has been noted within 24–48 hours of steroid treatment, surgery is indicated. Orbital decompression surgery may serve to relieve the pressure caused by swollen tissue.

Question 4
What differential diagnoses should be considered in a patient presenting with proptosis?

Differential diagnosis of proptosis include:
• orbital cellulitis (usually rapid onset and associated with fever)
• orbital tumours and metastases
• orbital pseudotumour, an idiopathic, nonspecific inflammatory orbital condition that is usually painful, progresses quickly compared to thyroid eye disease. It is almost always unilateral.
• carotid cavernous fistulas (patients present with proptosis, chemosis, diplopia and visual loss usually associated with a bruit)
• lesions or vascular accidents of the mid brain (can lead to restriction or paralysis of upgaze and pupil defects). CT and MRI scans can help lead to differentiation of these conditions.

CASE FOUR

Short case number: 4_23_4
Category: Nervous system & Eye
Discipline: Medicine
Setting: General practice
Topic: Systemic disease eye changes_autoimmune conditions

CASE
Felicity Masters [3_22_1], the 35 year old who was diagnosed with early rheumatoid arthritis ten years ago, now presents with a painful red left eye which she had noticed for the last few days. She thought it was conjunctivitis initially and had been using drops recommended by the local chemist.  **Significant history includes joint symptoms under control with no progression of deformity while on methotrexate and occasional oral steroids.** **On examination, the left eye was inflamed on the medial aspect with prominent conjunctival blood vessels.**

QUESTIONS
1. Outline the four possible eye diagnoses which are linked to rheumatoid arthritis. 2. How does episcleritis differ from scleritis in presentation and outcome?

Resources:

- **Oliva M and Taylor H. Conjunctival conditions. Australian Doctor, June 2004, p 39-46.**

ANSWERS

Question 1
Outline the four possible eye diagnoses which are linked to rheumatoid arthritis.

- Episcleritis is a relatively benign self-limiting condition, presumed to be an autoimmune process.
- Scleritis is a more serious condition which is often associated with an underlying connective tissue disease, most commonly rheumatoid arthritis.
- Infectious scleritis, typically with gram-negative bacteria, may occur after beta irradiation after pterygium removal. Infectious scleritis requires systemic antibiotic treatment.
- Dry eye is typically associated with a deficiency in aqueous tear production, known as kerato-conjunctivitis sicca. This condition may also be associated with systemic immune dysfunction from diseases such as rheumatoid arthritis and Sjögren's syndrome.

Question 2
How does episcleritis differ from scleritis in presentation and management?

Epiclertis	Scleritis
Episcleritis most often affects women aged 20-50. **Presentation**: Episcleritis generally has an acute onset and is characterised by a mild ache and focal areas of redness. Vision is normal and the patient may be asymptomatic. **Examination** shows a focal, nonraised, often well-defined patch of dilated episcleral vessels with underlying unaffected sclera. The condition may be confused with an inflamed pterygium. **Natural history**: Patients should be reassured that the condition does not threaten vision and should clear in several weeks. Prescription of an oral NSAID often assists with symptoms and may speed resolution. Artificial tears can also be prescribed for comfort. **Management**: Episcleritis that persists, worsens or frequently recurs should be referred. Topical corticosteroids may be considered by an ophthalmologist to speed resolution.	**Presentation**: With scleritis, patients complain of moderate to severe deep ocular pain, tearing and photophobia. **Examination**: The eye may be exquisitely tender to palpation. There is often brick-red inflammation of the sclera, episclera and conjunctiva that can be localised or diffuse. The underlying sclera may exhibit a bluish discoloration. The conjunctival blood vessels can be seen overlying the deeper inflammation. **Management**: Patients should be referred promptly to an ophthalmologist, who may investigate with a systemic workup. Treatment typically involves systemic corticosteroids or other forms of immune suppression.